Chris Cameron

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PhD researcher in electrophysiology (completion expected early 2027) and creator of Cubie. Part-time lecturer teaching undergraduate and postgraduate mechatronics, including computer architecture, C programming, circuit design, and embedded systems.

Cubie — GPU numerical integration

github.com/cubiepy/cubie

Built an open-source Python ODE solver for consumer NVIDIA GPUs to support electrophysiology simulations for a real-time application.

Draft, unpublished benchmarks show **10,000× speedup** over CPU implementations (SciPy, MATLAB) and **100×** over DiffEqGPU.jl.

- Optimised CUDA kernels using NVIDIA Nsight Compute: reordered operations, changed memory placement, and substituted individual instructions.
- Tuned occupancy to fit cache sizes and adjusted loop unrolling to fit the instruction cache.
- Instrumented internal solver stages to verify numerical correctness throughout the integration.
- Review AI-generated implementations through pull requests and agent sessions, identifying errors and validating numerical behaviour in a specialised codebase.

Open-source contributions

numba-cuda / numba-cuda-mlir

Merged and open pull requests addressing compiler performance, typing and lowering correctness, and AST transforms. Open pull requests for kernel performance controls.

SymPy — merged printer edit. **lcapy** — merged documentation edits.

Other projects

GPUODEBenchmarks github.com/ccam80/GPUODEBenchmarks

Performance benchmarks for GPU-based ODE solvers.

deep_zotero github.com/ccam80/deep-zotero

Local semantic search database for academic papers.